Name: ____________________ Problem Set 2 | Math 1180 | Cruz Godar

Due Thursday, September 17th at 11:59 PM

Complete the following problems and submit them as a PDF to Gradescope. You should show enough work that there is no question about the mathematical process used to obtain your answers, and so that your peers in the class could easily follow along. I encourage you to collaborate with your classmates, so long as you write up your solutions independently. Using AI tools to assist with the homework is not permitted. I write most of my problems to not require calculators, but you may find some of the problems require them — that's perfectly fine! All exam problems will be written without the need for calculators.

1. The following graph plots these four equations in some order:

   a) $z = \frac{1}{5}\left(x^3 + y^3\right)$.

   b) $z = xy$.

   c) $z = \sin(x) + \cos(y)$.

   d) $z = \sqrt{1 - x^2 - \left(\frac{y}{2}\right)^2}$.

Toggle each on using the expressions in the sidebar and match each function to its graph (you can technically do this by just typing the functions in as new expressions, but the point is to deduce it by analyzing the graphs).

2. Find an equation for a cylinder in $\mathbb{R}^3$ parallel to the y-axis, with radius 4, and whose central axis contains the point $(2, 3, 1)$.

3. Let $\vec{v} = \langle 0, 3, 4 \rangle$ and let $\vec{w} = \vec{i} - 2\vec{j} + \vec{k}$.

   a) Find $||\vec{v}||$ and $||\vec{w}||$.

b) Find $\vec{v} + \vec{w}$ and $2\vec{v} - \vec{w}$ in component form and sketch them both.

c) Find unit vectors in the same direction as $\vec{v}$ and $\vec{w}$, respectively.

4. Find an equation for a plane with normal vector $\langle 1, -2, 0 \rangle$ that passes through the point $\langle 2, 3, -8 \rangle$.

5. Find an equation for a plane that passes through $(0, 1, 1)$, $(-3, 1, 2)$, and $(1, 0, 2)$.

6. Find an equation for a plane that passes through $(1, 2, 3)$ and $(2, 2, 4)$ and is parallel to the y-axis.

In questions 7–9, determine whether the statement is true or false, where all of the vectors are in $\mathbb{R}^3$. If the statement is true, briefly justify why. If it is false, provide an example to demonstrate.

7. If $\vec{v} \bullet \vec{w} = 0$, then either $\vec{v} = \vec{0}$ or $\vec{w} = \vec{0}$.

8. If $\vec{v} \times \vec{w} = \vec{0}$, then either $\vec{v} = \vec{0}$ or $\vec{w} = \vec{0}$.

9. If $\vec{u} \bullet (\vec{v} \times \vec{w}) = 0$, then $\vec{u}$ is in the plane containing $\vec{v}$ and $\vec{w}$ when all three are drawn starting from the origin.

Let's try to find the general form of a unit vector in $\mathbb{R}^3$: rather than describing it as a vector $\vec{u} = \langle u_1, u_2, u_3 \rangle$ with $u_1^2 + u_2^2 + u_3^2 = 1$, it can be useful to have two parameters that we can freely vary instead of three parameters with a restriction. We'll find such a form in the next two questions.

10. Suppose $u_3 = 0$, so that if $\vec{u}$ is drawn with its tail at the origin, then the tip of $\vec{u}$ is in the xy-plane, as in the graph below. If θ is its angle measured counterclockwise from the positive x-axis, drawn in red, find a general formula for $\vec{u}$ in terms of its angle θ.

11. Now suppose $u_3 \neq 0$. We can think of $\vec{u}$ as being rotated up or down from the xy-plane by some angle φ; change the perspective on the graph and adjust φ_0 to see this in action. Hold θ fixed and look at the half-disk from the side, so that the problem effectively becomes 2D again. Use your answer to part a) and

this perspective to find an expression for $\vec{u}$ in terms of θ and φ.

Questions 12–16 concern the following situation: two companies, Aquilo and Fulgora, each produce steel and computer chips using labor. We can record a company's technology as a vector whose components are the numbers of labor hours needed to produce one unit of each good: Aquilo's is $\vec{a} = \langle 6, 3\rangle$, meaning six hours per unit of steel and three hours per unit of computer chips, and Fulgora's is $\vec{f} = \langle 1, 2\rangle$. A company has an **absolute advantage** in a good if it can produce a unit of that good in fewer hours. It turns out that a more useful notion is the **opportunity cost** of a good: the amount of the *other* good a company must give up in order to produce one more unit. We say that a company has a **comparative advantage** in a good if its opportunity cost for that good is lower than the other company's.

12. Which company has an absolute advantage in each good? Then find each company's opportunity cost of a unit of steel (your answers should be in units of computer chips), and do the same for computer chips. Which company has a comparative advantage in which good?

13. Now more generally, suppose $\vec{a} = \langle s_a, c_a, 0\rangle$ and $\vec{f} = \langle s_f, c_f, 0\rangle$, where all four of the component variables are positive. Then

$$\begin{aligned}\vec{a} \times \vec{f} &= \langle 0,\ 0,\ s_a c_f - c_a s_f\rangle \\ &= \left\langle 0,\ 0,\ c_f c_a \left(\frac{s_a}{c_a} - \frac{s_f}{c_f}\right)\right\rangle.\end{aligned}$$

Explain why the quantity in parentheses is the difference between the two companies' opportunity costs of steel. Since $c_f c_a > 0$, what does the *sign* of that quantity tell you? Check that it agrees with your answer to question 13.

14. Question 13 implies that $\vec{a} \times \vec{f} = \vec{0}$ exactly when the two companies have identical opportunity costs, in which case neither company has a comparative advantage in anything. When that is the case, what must be the angle θ between the two technology vectors?

15. Let's now look at specific numbers. Going back to the original vectors of $\vec{a} = \langle 6, 3\rangle$ and $\vec{f} = \langle 1, 2\rangle$, suppose each company has 12 labor hours available. First suppose each company splits its hours evenly between the two goods, and find the total world output of each good (assuming no other relevant companies

exist). Now suppose Fulgora spends all 12 of its hours on computer chips, while Aquilo spends only as many hours on computer chips as are needed to keep total world computer chips output unchanged, spending the rest on steel. Find the new world output. Who gained, and by how much?

16. How do you feel about this material and these questions? Is there anything in particular that you would like to focus on going forward?

17. Choose up to three questions (and at least one) on which you would like detailed feedback.